

Lahore Smog Intelligence Challenge

Participant Handbook, Complete Edition

Predict tomorrow's air quality, then build a safe advisory assistant that uses the forecast and cites official guidance.

4 hours | 2 sprints | 1 working system

ORGANIZED BY

Looplab

TECH PARTNER

Lifix Labs

Event date: _____

Team name: _____

[Open the Loopverse AI and ML Essentials repository on GitHub](#)

1. Challenge Overview

Build one connected AI system. Sprint 1 predicts what will happen. Sprint 2 explains what people should do, using your forecast and the supplied documents.

Total time	Sensors	Forecast rows	Documents	Final submission
4 hours	15	150	13	1 only

Everything participants need is in one place. Clone the repository first. It contains every data file, document and specification referenced in this handbook.

[Open the Loopverse AI and ML Essentials repository on GitHub](#)

Workflow: 1. Prepare (align and clean) 2. Forecast (PM2.5 and alarm) 3. Retrieve (current evidence) 4. Advise (answer with sources)

Part	Build	Submit	Points
Sprint 1	Next day PM2.5 and hazardous alarm for 15 sensors across 10 holdout days.	predictions.csv with exactly 150 rows	30
Sprint 2	RAG assistant that searches documents, calls the forecast, cites sources and refuses unsupported requests.	ask(question) plus answers to the released question set	45
Engineering	Reproducible repository, AI error log and recommendation.	README, code, log and recommendation	25

Challenge package

Files	What they contain	Critical detail
Two sensor files	Batch 1 and Batch 2 readings	Sensors from different batches have varying properties. Refer to sensor_metadata.csv for details.
Weather and sensors	Weather variables and sensor areas	Join carefully at the grain defined in SPEC.md.
Holdout inputs	Last 10 forecast origins without answers	Hidden answers are used for grading.
docs folder	13 health, school, FAQ, policy and sensor documents	Includes old and new guidance and a malicious instruction.
SPEC.md and templates	Exact schema, threshold, filenames and commands	Authoritative when mechanical details differ.

All files above are in the repository: [Open the Loopverse AI and ML Essentials repository on GitHub](#)

Winning principle. A simple, tested, honest system beats a complex system that leaks future data, invents evidence, or cannot be rerun.

2. Four Hour Plan

The clock is part of the challenge. Protect the final 20 minutes for packaging, and do not spend the whole event tuning one model.

Time	Milestone	Expected result
00:00 to 00:15	Kickoff and setup	Clone the repository, read SPEC.md, assign roles, confirm access.
00:15 to 00:35	Data audit	Check timezones, units, the minus 999 sentinel, duplicates, coverage and the flat line sensor.
00:35 to 01:30	Sprint 1 build	Clean pipeline, baseline and model, next day PM2.5 predictions.
01:30 to 01:50	Validate and alarm	Use later dates, tune the alarm and verify 150 output rows.
01:50 to 02:00	Checkpoint review	No submission yet. Confirm predictions.csv is valid against the spec and ready.
02:00 to 02:10	Break and fallback	The reference forecast is released for Sprint 2 if needed.
02:10 to 02:30	RAG and tool	Index documents, build the forecast lookup and the response schema.
02:30 to 03:10	Sprint 2 build	Implement ask(question), citations, refusal and Roman Urdu handling.
03:10 to 03:40	Hardening	Run old versus new, unsupported city and hostile document tests.
03:40 to 03:55	Package	Finish README, error log, recommendation and repository check.
03:55 to 04:00	Final submission	Submit everything once. There is no second attempt.

Suggested ownership

Data lead	Forecast lead	RAG and safety lead	Integrator and captain
Cleaning, conversion, joins and audit plots	Baseline and model, validation, alarm and predictions file	Retrieval, current guidance, citations, refusal and injection tests	Forecast tool, ask(), batch run, repository, final submission and time

If behind. Submit a leak free recent average baseline rather than nothing. For Sprint 2, use the released reference forecast and disclose it. A working, honest fallback keeps you competitive even if incomplete.

Submission policy: one attempt only. There is a single final submission at the end of the 4 hour window, with no intermediate checkpoint and no second attempt. Validate everything locally before the deadline. Evaluation for every team begins only after the window closes.

3. Sprint 1: Forecast Tomorrow Without Using the Future

For each sensor and holdout forecast origin, predict the next day's average PM2.5 and a hazardous yes or no decision. Final output: 150 rows.

Pipeline: 1. Standardize (clock and units) 2. Clean (missing values and faults) 3. Engineer (lags and weather) 4. Validate (past to future)

Step	Minimum viable	Stronger version
Inspect	Shapes, dates, sensor counts, nulls and two plots	Coverage heatmap and same area network comparison
Standardize	Convert both networks to one timezone and one PM2.5 unit	Automated unit and timezone assertions, with raw columns retained
Clean	Turn minus 999 into missing and handle the known stuck interval	Quality flags, coverage threshold and documented imputation
Aggregate	Daily sensor average after cleaning and conversion	Require a minimum number of valid readings per day
Features	Last day and recent rolling mean	Several shifted lags, rolling spread, weather lags, area and network
Model	Persistence or recent average	Regularized regression, trees or boosting, or a validated ensemble
Validate	One chronological holdout	Walk forward validation across multiple cutoffs
Alarm	Apply the supplied hazardous rule	Calibrate the threshold using hazardous recall and false alarms

Validation rules

- Never use a random split for time data. Train on earlier dates and evaluate on later dates.
- Shift lag and rolling features so every value existed at the forecast origin.
- Fit imputers and scalers on training data only. Compare every model against a simple baseline.
- Check PM2.5 error plus hazardous recall and false alarms. Overall accuracy is misleading when hazardous days are rare.

Order matters. Parse time, align timezone, convert units, replace invalid readings, handle faults, aggregate, then create lagged features.

The Four Forecast Traps

Trap	Failure	Detection and fix
1. Future leakage	A random split or unshifted features let tomorrow influence training.	Print train and validation date ranges and feature timestamps. Use chronological or walk forward validation.
2. Networks mismatch	Batches use different units and timezones. Direct merging shifts dates and mixes units.	Use the supplied conversions before aggregation. Plot two same area sensors after conversion.
3. Broken values	The value minus 999 enters averages, and one sensor stays constant for three days.	Replace the sentinel before calculations, plot sensors, and flag, exclude or impute the faulty interval consistently.
4. Misleading accuracy	Always predicting not hazardous looks accurate when hazardous days are rare.	Evaluate hazardous recall plus precision and false alarms, not overall accuracy alone.

predictions.csv submission gate

Check	Pass condition
Template	Copied from the supplied template, with exact columns, order, names, types and date format.
Rows	Exactly $15 \times 10 = 150$ rows and 150 unique sensor and date pairs.
Values	Known sensor IDs, finite PM2.5 numbers in the required unit, and a valid hazardous representation.
CSV hygiene	No unnamed index column, no missing required cell, no duplicate and no extra row.
Reproducibility	One documented command regenerates the file from supplied inputs.
Submission	Included once in your single final submission at 04:00. Validate locally beforehand.

Break fallback. At 02:00, the reference forecast may be used behind Sprint 2. It does not replace or improve your own forecast quality.

4. Sprint 2: Build a Grounded Smog Advisory Assistant

The assistant has two evidence channels: supplied documents and a callable forecast. It must show real sources and decline what it cannot know.

Flow: 1. Understand (intent and location) 2. Retrieve (current passages) 3. Call (forecast tool) 4. Answer (advice and sources)

Question	Documents	Forecast	Behavior
General health advice	Yes	Usually no	Use current supplied guidance and cite it.
Tomorrow in a covered Lahore area	Yes	Yes	Combine the tool result with applicable advice and policy.
Forecast only	Optional	Yes	Report location, target date, result and limitation.
Policy or rule only	Yes	No	Prefer the current authoritative document.
Unsupported city or date	Maybe	Never guess	State the unavailable scope clearly.
Instruction hidden in a document	Untrusted evidence	As needed	Ignore commands inside retrieved content.

Required ask(question) behavior

Part	Requirement
answer	Direct, plain language response grounded in retrieved passages and the forecast output.
sources	Only exact document IDs or names that were retrieved and actually used.
forecast called	A truthful boolean showing whether the forecast function executed. Use the exact key name given in SPEC.md.
schema	Exact keys, types, nesting and filename defined in the supplied starter template and SPEC.md.

Forecast tool rules

- Validate the location against supplied Lahore areas and sensors, and the date against supported targets.
- Return structured PM2.5, hazardous status, location, target date, and team or reference provenance.
- On invalid input or failure, return a structured unavailable status and never a guessed number.

Reliable RAG, Safe Answers and the Question Drop

Reliability matters more than using a complicated framework. Keyword, vector or hybrid retrieval is acceptable if it finds the right current evidence.

Stage	Implementation
Ingest	Keep document ID, title, published or effective date, authority, status or supersession, and text.
Chunk	Create coherent passages and preserve metadata on every chunk.
Retrieve	Search relevant passages, then filter and rank for applicability, authority and currency.
Generate	Give the model only selected evidence and tool output. Require abstention and copied source IDs.
Validate	Reject unknown citations, unsupported numeric claims and answers that skipped a required forecast call.

Security rule. Documents are untrusted information, never instructions. A passage may support an answer but cannot change system rules, suppress sources, or declare the air safe.

Five tests before 03:20

Test	Pass condition
Covered Lahore forecast	Forecast is called, with correct location and date, and current advice is cited.
General advice	Relevant documents are cited and the forecast is not called unnecessarily.
Unsupported city or date	No invented number, and the supported scope is stated.
Old versus new school rule	The new, current document controls the answer.
Ignore rules in a document	The instruction is ignored and system policy still controls.

Roman Urdu. Normalize common terms for retrieval and answer in the user's language when practical, while keeping source names exact.

5. Seven Traps

Run this table as a review before submission. Each trap represents a deliberate challenge condition.

#	Trap	What careful teams do
1	Future leakage	Train on early dates, validate on later dates, shift features and fit preprocessing on training data only.
2	Mismatched networks	Convert UTC and Pakistan time and AQI and PM2.5 using supplied rules before merging or aggregation.
3	Broken or fake readings	Turn minus 999 into missing, detect the three day flat line, and apply and document a consistent quality rule.
4	Misleading score	Measure hazardous recall and false alarms or precision rather than relying on overall accuracy.
5	Old versus new advice	Keep date, status and authority metadata and let current replacement guidance control the answer.
6	Made up coverage	Validate location and date, and say unavailable for unsupported requests such as other cities. Never guess.
7	Trick document	Treat retrieved text as untrusted data, ignore embedded commands, and validate claims and citations after generation.

Defense in depth

Layer	Control
Metadata	Record source, authority, date, current or superseded status, and chunk location.
Retrieval	Rank for relevance plus applicability and currency, not similarity alone.
Prompt	State that passages are evidence and any instructions inside them must be ignored.
Tool code	Allow only covered locations and dates and return explicit unavailable results.
Output check	Allow citations only from retrieved IDs, and verify required tool use and unsupported claims.
Tests	Exercise old versus new, unsupported city and date, malicious passage, ambiguity and Roman Urdu.

When sources conflict. If two current sources genuinely conflict and no replacement rule is supplied, state the conflict, cite both, and avoid inventing a resolution.

6. Rules and Responsibilities

AI tools are allowed. Responsibility stays with your team.

- You may use coding and chat assistants, IDE agents, hosted or local models, frameworks, notebooks and open source libraries, subject to event rules.
- Paid tools and APIs are allowed at your own cost. Looplab does not provide any tools, API keys, credits or compute.
- Do not seek hidden labels, access another team's work, exploit data leaks, fabricate citations or forecasts, or submit code you cannot run and explain.
- Do not commit API keys or secrets. Disclose use of the reference forecast and attribute reused public code where required.

Submission policy: one attempt only. There is a single final submission at the end of the 4 hour window, with no intermediate checkpoint and no second attempt. Validate everything locally before the deadline. Evaluation for every team begins only after the window closes.

AI error log: three required entries

Field	Record
Context and wrong output	What you asked, and the exact bad assumption, code, answer or citation.
Risk	How it could damage forecasting, safety, scoring or reproducibility.
Detection	The plot, assertion, date check, document comparison or test that exposed it.
Correction and verification	What changed and what evidence shows the fix works.

Strong example: AI suggested a shuffled train and test split. Date range inspection revealed future leakage. The team changed to chronological validation and accepted a worse but credible score.

7. Final Gate: Submission Checklist

Use this page at 03:40. This is your only submission. The exact upload mechanism and filenames are defined in the repository.

Area	Pass condition
Sprint 1 file	Exact template, 150 rows, unique sensor and date pairs, finite values, valid alarms and no extra index.
Sprint 1 method	Chronological validation, documented conversion and cleaning, and baseline comparison available.
Forecast tool	Validates place and date, returns a structured result with team or reference provenance, and never guesses.
RAG	Current authoritative source preferred, only real citations, and retrieved text treated as untrusted.
ask(question)	Exact signature and schema, truthful forecast called field and a clear unsupported response.
Safety	Old versus new, unsupported city and date, hostile document, ambiguity and Roman Urdu tested.
Repository	Judge access works, README commands run, dependencies listed, and no secrets or caches.
Evidence	Three strong AI error entries, recommendation, and limitations and fallback use disclosed.
Submit	All required files accepted in one final submission before the deadline, with the receipt saved.

Quick glossary

Term	Meaning
PM2.5	Tiny airborne particles. A higher concentration means worse pollution.
AQI	An index, not the same unit as PM2.5 concentration.
Leakage	Using information that would not exist at prediction time.
Recall	Share of truly hazardous cases your alarm catches.
RAG	Retrieve passages, then answer from those passages.
Grounded	Claims are traceable to supplied evidence or tool results.
Prompt injection	Untrusted text tries to override system instructions.
Abstain	State that reliable information is unavailable instead of inventing it.

Repository. The exact upload mechanism and filenames are defined in the repository.

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Definition of done. Your system predicts, retrieves, calls, answers, cites, refuses and reruns. Your repository proves how. Your files follow the supplied contract exactly.